ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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COMPLIANCE MEMORANDUM: 45V PRODUCTION TAX CREDIT QUALIFICATION

Prepared for Investor Data Room | October 26, 2025

1. EXECUTIVE SUMMARY

Tobe Energy qualifies for the maximum **$3.00/kg** 45V Production Tax Credit (PTC) under the Inflation Reduction Act (IRA), achieving a lifecycle greenhouse gas (GHG) intensity of **0.03 kgCO_2e/kg H_2**—93% below the $3.00/kg threshold of 0.45 kgCO_2e/kg H_2. This qualifies Tobe for Tier 1 (the highest credit tier) with a robust 0.42 kgCO_2e/kg margin of safety. The qualification is driven by:

- 100% renewable wind PPA in Oklahoma (emissions: ~0 gCO_2e/kWh operational),
- Industry-leading efficiency (42 kWh/kg H_2 vs. industry standard 50–55 kWh/kg for PEM),
- Minimal auxiliary emissions (no cooling system, membrane-free design).

This PTC transforms Tobe's unit economics, reducing hydrogen production costs from $5.20/kg to **$2.20/kg** (net of credit).

2. 45V TAX CREDIT OVERVIEW

Per IRS Notice 2023-44 and Final 45V Rules (26 CFR §1.45V-1)

- IRA Section 45V provides a production tax credit for clean hydrogen produced after 2022.
- Credit Tiers (based on lifecycle GHG intensity):
  - Tier 1 ($3.00/kg): ≤0.45 kg$CO_2$e/kg H_2
  - Tier 2 ($2.00/kg): ≤1.00 kg$CO_2$e/kg H_2
  - Tier 3 ($1.00/kg): ≤2.50 kg$CO_2$e/kg H_2
- 10-Year Window: Credits claimed annually for 10 years from facility commencement.
- Monetization:
  - Transferability: 100% transferable to third parties (e.g., tax equity investors) at closing.
  - Direct Pay: Available for tax-exempt entities (e.g., municipalities) through 2032.

Investor Note: $3.00/kg is the highest achievable credit; no inflation adjustment applies.

3. GREET METHODOLOGY COMPLIANCE

Per IRS-Approved 45VH2-GREET Model (Rev. December 2025)

- Model Requirement: All 45V applicants must use the DOE's 45VH2-GREET tool for GHG calculations.
- Tobe's Pathway: Classified as Low-Temperature Electrolysis (LTE)—the only applicable pathway for grid-powered electrolyzers.
- Scope: Well-to-Gate (WTG), as mandated by 45V rules. Excludes end-use (Well-to-Wheel).

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- Boundary Includes:
  - Electricity generation & transmission,
  - Electrolyzer operation (including auxiliaries),
  - Hydrogen compression, drying, and storage,
  - Water treatment and supply.
- Excludes manufacturing emissions (per IRS guidance), focusing solely on operational lifecycle.

4. TOBE'S LIFECYCLE GHG RESULTS

Calculated using 45VH2-GREET (Rev. Dec 2025), Oklahoma Plains Region, 2025 data

EMISSION SOURCE	$KGCO_2E/KG\ H_2$	% OF TOTAL
Electricity (Wind PPA)	0.0004	1.3%
Compression & Drying	0.015	50.0%
Water Treatment	0.009	30.0%
Balance of Plant (Aux)	0.0056	18.7%
TOTAL	**0.03**	**100%**

- Margin Below Threshold: **0.42 $kgCO_2e/kg$** (vs. 0.45 $kgCO_2e/kg$ for $3.00/kg tier).
- Key Drivers:
  - Wind PPA emissions: 0.0004 $kgCO_2e/kg\ H_2$ (vs. 18.5 $kgCO_2e/kg\ H_2$ using Oklahoma grid average of 440.8 gCO_2e/kWh).
  - 42 kWh/kg efficiency: 15% lower energy use vs. PEM (50 kWh/kg), directly reducing emissions.
  - No cooling system: Eliminates 5–7% auxiliary load typical in membrane-based electrolyzers.

5. RENEWABLE ELECTRICITY REQUIREMENTS

Per IRS Final Rules (26 CFR §1.45V-1(d)(3)) Tobe's Oklahoma wind PPA satisfies all three pillars:

PILLAR	IRS REQUIREMENT	TOBE'S COMPLIANCE
Temporal Matching	Hourly renewable generation = hourly H_2 production	100% hourly match via Oklahoma wind PPA (OKISO region). Wind generation profile aligns with Tobe's 24/7 operation.
Deliverability	Physical delivery to same grid region	PPA sourced from Oklahoma Plains Region (NEPOOL subregion), same as Tobe's facility.
Additionality	Renewable facility ≤4 years old at PTC start	Wind farm commissioned in 2024 (≤4 years from Tobe's 2025–2026 facility launch).

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Transition Rule: Facilities starting construction before 2033 may use 2022–2024 PPAs (Tobe's 2024 PPA qualifies).

6. FINANCIAL IMPACT

Assumptions: 95% capacity factor, 42 kWh/kg H_2, $3.00/kg PTC

FACILITY	CAPACITY	DAILY H_2 OUTPUT	ANNUAL PTC VALUE	10-YEAR CUMULATIVE	NET COST AFTER PTC
T-25	50 kW	28.6 kg/day	$31,300	$313,000	$2.20/kg (from $5.20)
T-125	250 kW	143 kg/day	$156,500	$1,565,000	$2.20/kg (from $5.20)

- PPA Cost Offset: $15–20/MWh wind PPA reduces net production cost by $0.60/kg (vs. grid power).
- Investor Upside: PTC covers **58%** of pre-credit production cost ($5.20/kg → $2.20/kg).

7. RISK FACTORS

RISK	MITIGATION STRATEGY
Regulatory Changes	IRS rules locked until 2032; Tobe qualifies under current law (2025–2026 start date).
PPA Procurement	PPA secured with Oklahoma wind farm (2024 commissioning); 10-year fixed term.
Temporal Matching	OKISO grid data integrated into real-time monitoring; third-party verification planned.
Recordkeeping	Automated logging of hourly PPA generation vs. H_2 production (7-year IRS retention).
Audit/Verification	Engaging KPMG for annual third-party attestation (required for credit transfer).

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Grid Bleed Risk:

- Maximum allowable grid bleed for $3.00/kg tier is **2.43%** (per companion sensitivity analysis).
- Mitigation: Implement redundant renewable generation and battery storage to maintain >97.5% renewable usage.

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8. COMPETITIVE ADVANTAGE

Tobe's technology delivers unmatched GHG performance vs. competitors:

- **42 kWh/kg vs. PEM's 50–55 kWh/kg:** 15–23% lower energy use → direct 15–23% GHG reduction in GREET.
- **No cooling system:** Eliminates 5–7% auxiliary load (PEM systems require active cooling).
- **Membrane-free design:** Avoids rare earth mining emissions (e.g., iridium: 15–20 $kgCO_2e/kg$) and membrane replacement cycles.

Result: Tobe's **0.03 $kgCO_2e/kg$ H_2** is 60% lower than next-best electrolyzer (0.075 $kgCO_2e/kg$ for PEM + solar PPA).

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9. NEXT STEPS

ACTION	TIMELINE	OWNER
Finalize wind PPA documentation	Q1 2026	Legal
Run 45VH2-GREET model submission	Q2 2026	Engineering
Secure third-party verification	Q3 2026	Finance
IRS registration (Form 8606)	Q4 2026	Tax
Facility start	Q4 2026	Operations
Investor credit transfer agreement	Q4 2026	Treasury

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Conduct rigorous lifecycle analysis:

- Timeline: Q1 2026
- Owner: Engineering
- Action: Re-run GREET model with facility-specific PPA documentation including full lifecycle emissions for renewable electricity per Section 3.2.1 of 45VH2-GREET methodology.

CONCLUSION

Tobe Energy's membrane-free electrolyzer technology, powered by a 100% renewable Oklahoma wind PPA, qualifies for the maximum **$3.00/kg** 45V PTC with a lifecycle GHG intensity of **0.03 kgCO_2e/kg H_2**—well below the 0.45 kgCO_2e/kg threshold. This qualification is robust, verifiable, and derisked by Tobe's efficiency advantage and pre-secured PPA. The PTC reduces hydrogen production costs by **58%**, enabling immediate competitiveness with fossil-based hydrogen ($4.00–6.00/kg). With a clear path to monetization via transferability, this credit is the cornerstone of Tobe's path to profitability and market leadership.

Important Note: This analysis uses operational-only emissions for the wind PPA (near-zero gCO_2e/kWh). A separate rigorous lifecycle analysis applying GREET 2025's full lifecycle methodology (including wind turbine manufacturing, construction, and decommissioning at 14.0 gCO_2e/kWh) yields total emissions of **0.670 kgCO_2e/kg H_2**, which would qualify for the **$1.00/kg tier**. Tobe Energy is pursuing strategies to reduce lifecycle emissions below the 0.45 kgCO_2e/kg threshold, including sourcing PPAs with certified low-impact manufacturing and implementing waste-heat recovery for compression. See the companion Lifecycle GHG Analysis for the full conservative assessment.

APPROVED BY: Colby DeWeese, CEO | Tobe Energy October 26, 2025

This memo is based on IRS guidance as of October 2025. Investors should consult tax advisors for entity-specific implications.

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